SearchScape

Project Idea: -

SearchScape: Interactive Simulation and Comparative Analysis of AI Search Algorithms in Grid-Based Environments

1. Introduction: -

Artificial Intelligence search algorithms are fundamental techniques used to solve pathfinding and decision-making problems.

This project presents **SearchScape**, an interactive educational platform that visualizes how different search algorithms explore a grid-based environment to find an optimal path from a start node to a goal node.

Unlike traditional static visualizations, the system provides a **real-time simulation** where an intelligent agent moves step-by-step, making decisions dynamically and adapting to obstacles.

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2. Problem Statement: -

Pathfinding in environments with obstacles is a classical AI problem. The challenge is to find an optimal path between a start and a goal node while minimizing cost, time, and explored nodes.

This project aims to:

- Simulate realistic navigation behavior
- Compare multiple search algorithms
- Analyze performance under different conditions
- Provide an educational visualization tool

3. Objectives:-

- Develop an interactive grid-based simulation system
- Allow full user control over the environment
- Implement multiple AI search algorithms
- Provide real-time agent movement simulation
- Compare algorithm performance using metrics
- Build an educational platform for learning AI concepts

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4. System Features:-

4.1 User-Controlled Environment

The user has full control over the grid:

- Adjustable grid size (10×10 to 40×40)
- Manual selection of:
  - Start node (agent position)
  - Goal node (destination)
- Obstacle creation:
  - Manual drawing (by click)

4.2 Simulation System

- The agent moves step-by-step
- Makes dynamic decisions
- Performs backtracking when necessary
- Mimics real-world navigation behavior

4.3 Visualization

- Start → boy character
- Goal → Treasure character
- Obstacles → Realistic Tree
- Visited nodes → Pink
- Final path → Blue

5. Algorithm Used: -

Uninformed Search

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Uniform Cost Search (UCS)
- Depth-Limited Search (DLS)
- Iterative Deepening Search (IDS)
- Bidirectional Search (BS)

Informed Search

- Best-First Search (Greedy Search)
- A* Search

Heuristics for A*

- Manhattan Distance
- Euclidean Distance

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6. Performance Matrics: -

After each execution, the system calculates:

- Path Length
- Execution Time
- Number of Nodes Explored

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7. Comparative Analysis: -

The system generates a detailed report comparing all algorithms.

7.1 Tabular Comparison

Algorithm	Path Length	Execution Time	Nodes Explored	Efficiency

7.2 Insights

- Fastest algorithm
- Most efficient algorithm

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8. Educational Features

- Step-by-step execution mode
- Comparison page

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9. PEAS Analysis: -

Element	Description
Performance	Path Length, Nodes Explored, Execution Time
Environment	Grid-based environment with obstacles
Actuators	Agent movement and visualization
Sensors	Grid state, node detection

10. ODESDA Environment Classification: -

Property	Type
Observability	Fully Observable
Determinism	Deterministic
Episodic / Sequential	Sequential
Static / Dynamic	Static
Discrete / Continuous	Discrete
Agent Type	Single-Agent

11. Agent Type: -

The system is modeled as a **Goal-Based Agent**, as it operates with a clear objective of reaching a goal node from a start node using search and planning strategies.

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12. Final Summary: -

SearchScape is an interactive AI educational platform that transforms abstract search algorithms into a visual, real-time, and user-controlled simulation environment. It bridges the gap between theory and practice by allowing users to design environments, run algorithms, and analyze performance results.